

CLASS TEST.

(1) Architectural patterns and MVC.

→ Architectural patterns are reusable structures that define how the components of a software system are organized and interact.

Two common patterns:-

→ MVC (Model - view - Controller)

→ Layered architecture.

MVC improves modularity and maintainability by separating:-

→ Model - data and business logic

→ view - user interface

→ Controller - handle user interface.

(2) DAO pattern:-

→ DAO (Data access object) is a design pattern that separates database operations from applications / business logic.

Advantages:-

→ Makes database code easier to maintain

→ Allows the database implementation to be changed easily.

Example:-

STUDENTDAO handles operation such as insert student(), update student(), get student(). while the application logic does not directly interact with SQL.

③ Object oriented design guidelines and LSP:-

→ Object oriented design guidelines are principle used to create software that is reusable, maintainable and flexible.

Two categories:-

- ① Class-level guidelines:- focus on designing individual classes.
- ② Relationship-level guidelines:- focus on relationship between classes.

Liskov Substitution principle:- A subclass should be able to replace its superclass without causing incorrect behaviors.
ex:- a car is a subclass of a vehicle.

④ Database persistence and ORM:-

→ Database persistence means storing an object's data permanently so that it remains available after the application ends.

Two persistence mechanism:-

- ① Object-relational mapping (ORM)
- ② Serialization / file storage.

ORM:-

→ ORM framework map objects / classes to database tables. They allow programmers to work with objects instead of writing SQL for every database operation.

⑤ Application Layer:-

→ The Application layer coordinates application use cases and controls the flow of operation.

Two responsibilities:-

- ① coordinates use cases
- ② Manage communication between presentation and domain layers.

→ It does not directly implement business rules. Instead, it calls appropriate domain object to perform rules.

⑥ Three-tier Architecture:-

- ① Presentation layer - user interface
- ② Business layer - business processing and rules
- ③ Data layer - database operation.

E-commerce transaction flow.

Customer → presentation → Business → Data → Database.

⑦ Business logic layer:-

→ Business logic layer contains and manages the rules and processing of a application.

Two functions:-

- ① validate request
- ② process business operation

For example:- in an online shopping system, it checks whether the product is available and then calculates the order total before sending the result back to presentation layer.

⑧ Cohesion and coupling:-

→ cohesion refers to how closely related the responsibilities of a class/module are

→ coupling refers to the dependency between different classes/modules.

Good design aims for:-

High cohesion + low coupling. (This makes software easier to maintain, modify, test).

⑨ Encapsulation and Data Hiding

→ Encapsulation means combining data and methods inside a class and controlling access to the data

ex:- class bankaccount {

private:

double balance;

public:

void deposit (double amount) {

balance += amount;

}

};

→ data hiding specifically focuses on preventing direct access to internal data.

⑩ Layered online banking system suitable architecture is

Presentation → Application → Business → Data → Database.

- ① Presentation:- Login and banking interface
- ② Application: coordinates transaction
- ③ Business:- validates transfers and account rules
- ④ Data:- Handles database access.

Suitable patterns:-

- MVC for presentation
- DAO for database
- Singleton for carefully controlled shared resources.

⑪ Infrastructure layer:-

→ The infrastructure layer provides technical services required by other application layers.

Two responsibilities:-

- ① Database access
- ② Communication with external system / service.

⑫ Singleton Design patterns:-

→ Singleton is a design pattern that ensures a class has only one instance and provides a common access point to it.

Characteristics:-

- only one object is created
- provides global / shared access.
- controls object creation.

⑬ Dependency injection.

→ Dependency injection (DI) is a technique where an object's required dependencies are provided from outside instead of the object creating them self.

Benefits :-

→ Reduce tight coupling.

→ Improves testability.

⑭ Strategy Design pattern :-

→ The strategy pattern allows different algorithm/behaviours / beh. to be selected at runtime.

It has 2 main roles :-

→ Strategy :- Define the common interface for different algorithm.

→ Context :- Uses the selected strategy without knowing its internal implementation.

⑮ Library Management system :-

layered architecture :-

Presentation → Application → Business → Data → Database.

Layers :-

→ Librarian / student interface

→ Application :- coordinates book issue / return operation.

→ Business :- checks availability, borrowing rules.

→ Data :- performs database operations.